

CLINICAL EXAMINATION



OF

THE ABDOMEN

OUTLINE

- Revision on Topographical Division of the Abdomen.
- Pre – requisites for Abdominal Examination
- Components of Abdominal Examination
 - The Four Cardinal steps of Abdominal Examination.
 - Differential Diagnoses for Abnormal Findings on Examination.
 - Reporting of Examination Findings.

1) TOPOGRAPHICAL DIVISION OF ABDOMEN

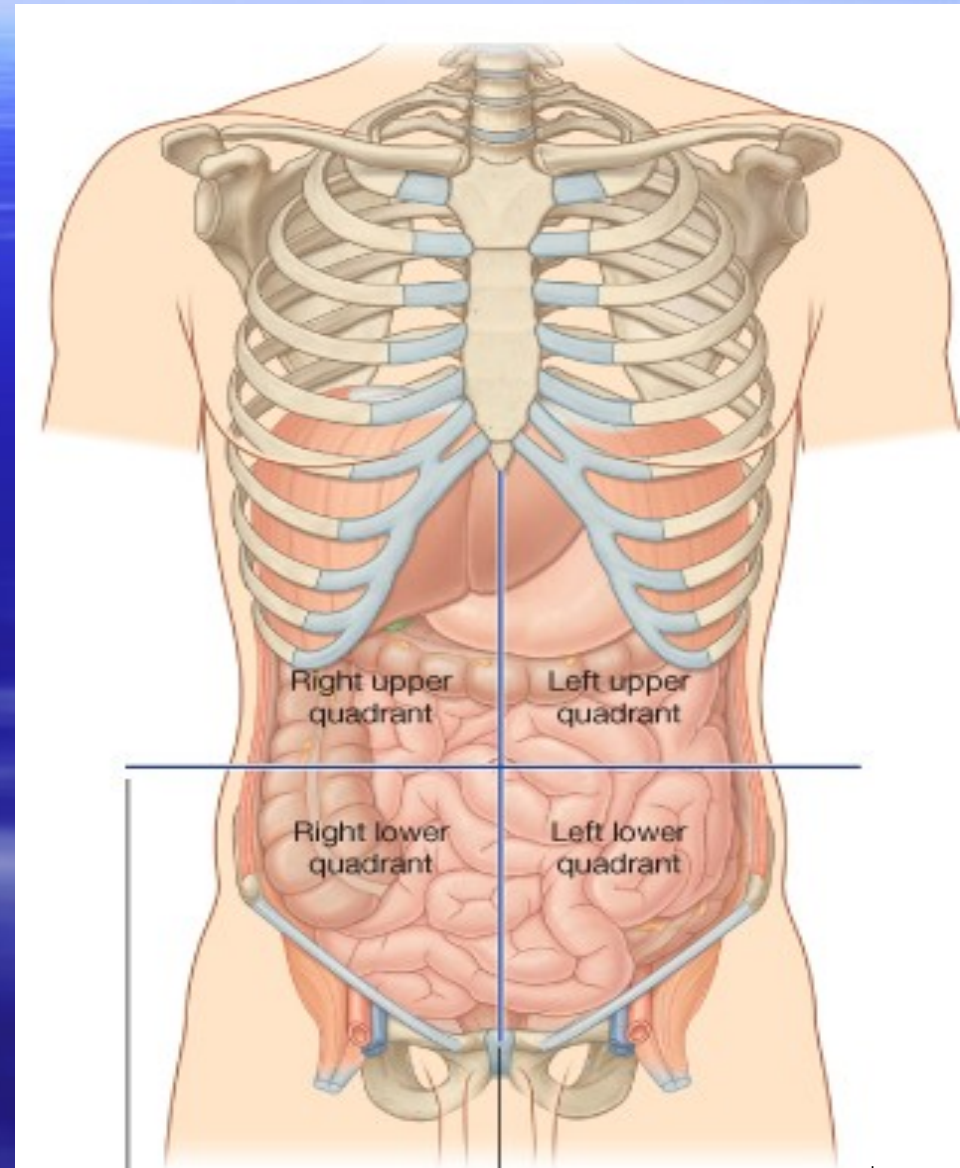
- Used to describe the location of abdominal organs.
- There are two patterns of topographic division.
 1. Four quadrant pattern.
 - it visualizes abdomen in 4 regions.
 2. Nine region organization pattern

1. Four quadrant pattern

- It is made using two imaginary planes.

1.A **Horizontal** Plane :- the trans umbilical plane that passes through umbilicus and the inter vertebral disc between L3 & L4.

2.A **Vertical** plane :-the trans umbilical median plane from Xiphisternm to pubic angle.



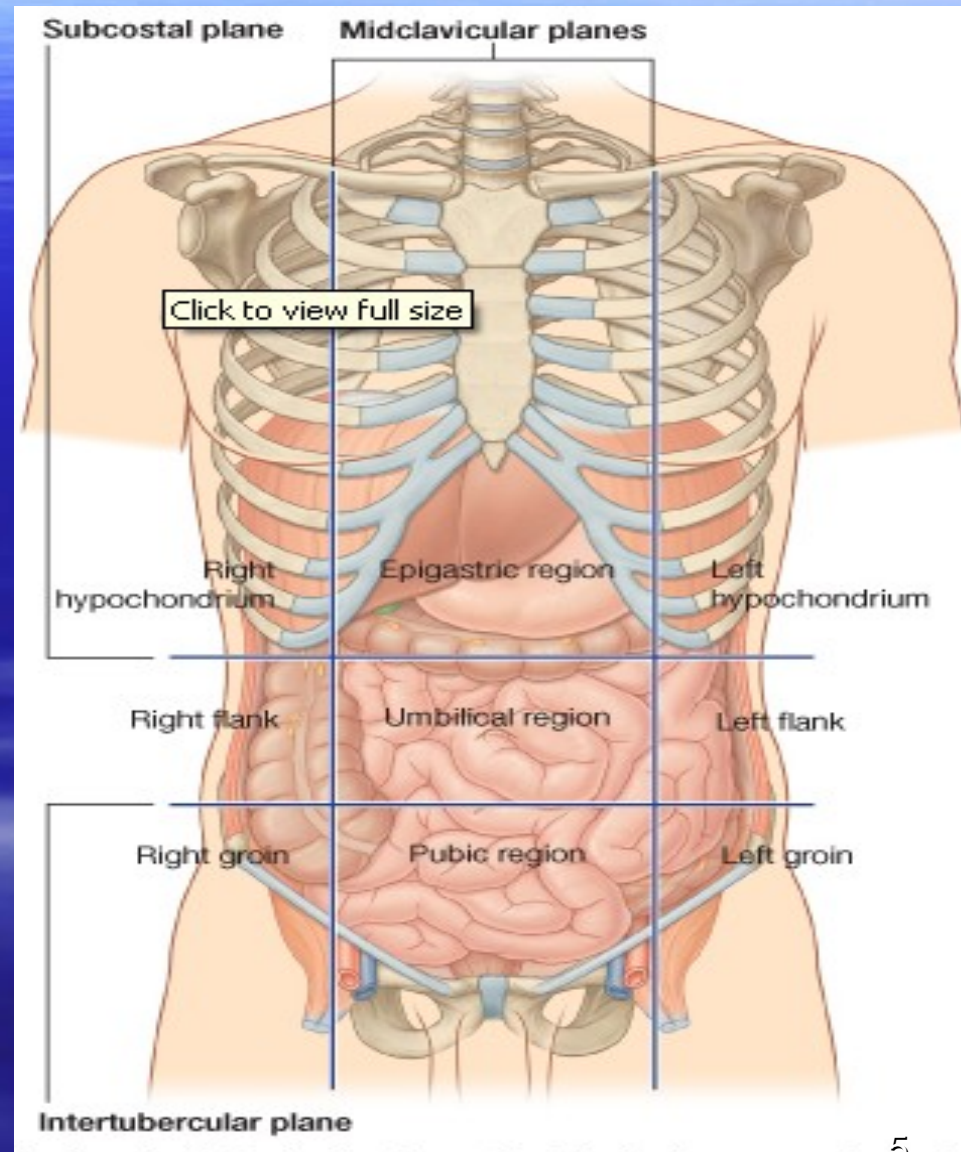
2. Nine region organization pattern

- made using 2 imaginary Horizontal & 2 Vertical Planes.

1. **Horizontal** Planes:- are

- Sub costal plane &
- Inter tubercular plane.

2. **Two Vertical** planes :-
passes from mid point of clavicle inferiorly to a point midway b/n ASIS & pubic symphysis.



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Prerequisites for Abdominal Examination

1. Good Light
2. Relaxed patient
3. Position : supine, arms at sides or folded across.
3. Full exposure from above the xiphoid process to symphysis pubis.
 - Groin should be visible.
 - Genitalia should be draped



Components of Abdominal examination

1. The Four Cardinal Steps.

Can be done in two different orders.

A

- Inspection
- Auscultation
- Palpation
- Percussion

2. Digital Rectal Examination.



1. INSPECTION

- Stand on - the patient's right side+/- foot end.

-INSPECT For:-

1. **SHAPE** of Abdomen .
(Contour, Symmetry & Flank fullness)

- Normal – Flat
- Abnormal Shapes

<u>Shape</u>	<u>Cause</u>
*Scaphiod	-Cachexia
*Distended	- 5 F's Fat, Fluid
Generalized	- Fetus, Flatus or Feces.
Localized (bulges)	- Mass (tumor), - Organomegally.



2.MOVEMENT of Abdominal wall

- Normally. Abdomen,
 - rise – during Inspiration.
 - fall – during Expiration.
- Abnormal.
 - Silent (No Mov't)
 - In Generalised Peritonitis

3.VISIBLE Peristaltic Movement

- Normally.
 - Not Visible
- Abnormal
 - Visible - In Pyloric stenosis & Dynamic BO.

4. SKIN and surface of the Abdomen

- Skin color & Luster
- Look for presence of:-
 - * Scars - old (White)
 - New (Pink)
- Lineanigra- in pregnant's Abdomen.

Cont'd..

- *Striae
 - white – Old
 - Pink – recent
 - Purple – In cushing syndrome.

*Rashes & skin lesions

5. **Dilated** Superficial Ab dominal wall veins

- Look for direction of flow (refilling Pattern)

<u>Causes</u>	<u>Direction of flow</u>
- Portal HTN	- up ward
- IVC- obstn	- down ward

6. **Umblicus**

Normally – slightly retracted & inverted

7. **Hernia**:- inspect Epigasric, Umblical ,Femoral & Surgical sites



2. Auscultation

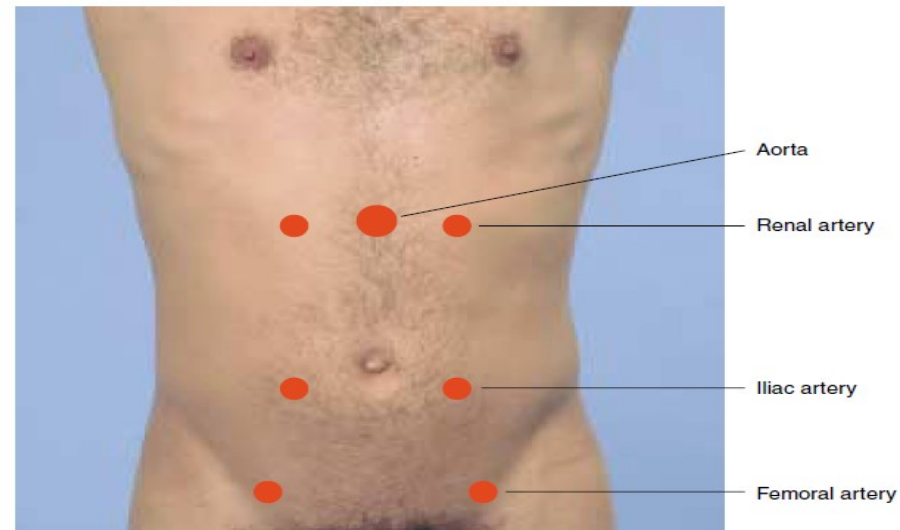
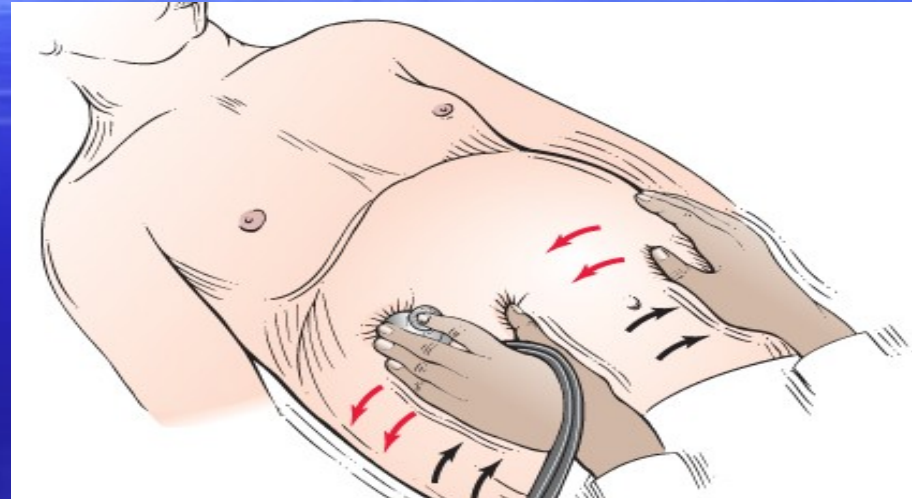
Purpose:-

1 To detect bowel Sound
- at mid line of Abdomen.

- Normal 5-30 clicks & Gurgle sound/ 1mt
- Hyper/ Hypo- $>30/ <5$
- Absent – No Sound/2mts

2. To detect obstructed Viscus. (Succesion Splash)

3 To detect Abdominal aa stenosis (bruit)





■ 3. PALPATION

1. Prerequisites for Palpation.

- Patient Should have empty bladder.
- Warm your Hands Before palpation.
- Avoid long finger nails.

2. Rules of Palpation.

- Approach slowly & avoid unexpected movement.
- Ask the patient to point areas of pain & examine this area at last.
- Watch patient's face closely for sign of Tenderness.
- Distract the patient with conversation.

3.Steps in palpation

1. Superficial (Light) palpation.
2. Deep palpation.
- 3.Liver palpation
- 4.Spleen palpation
- 5.Kidneys palpation
- 6.Balloting (Identifying an organ or Mass in Ascitic Abdomen)

1. Superficial(Light)palpation



- **Purpose:** used to detect superficial mass, tenderness & Guarding (voluntary/Involuntary Spasm of Abd.m.m.
- **Technique** - palpate with flat part of hand not finger tips
 - Hand should be molded / lifted from area to area.
 - Always start from area farthest to point of pain.
 - usually start from RLQ/LLQ ,& proceed CW/ CCW.
- ❑ Abnormal finding.
 - Diffuse guarding - indicates Generalized peritonitis. (Board like Abdomen)

2. Deep palpation



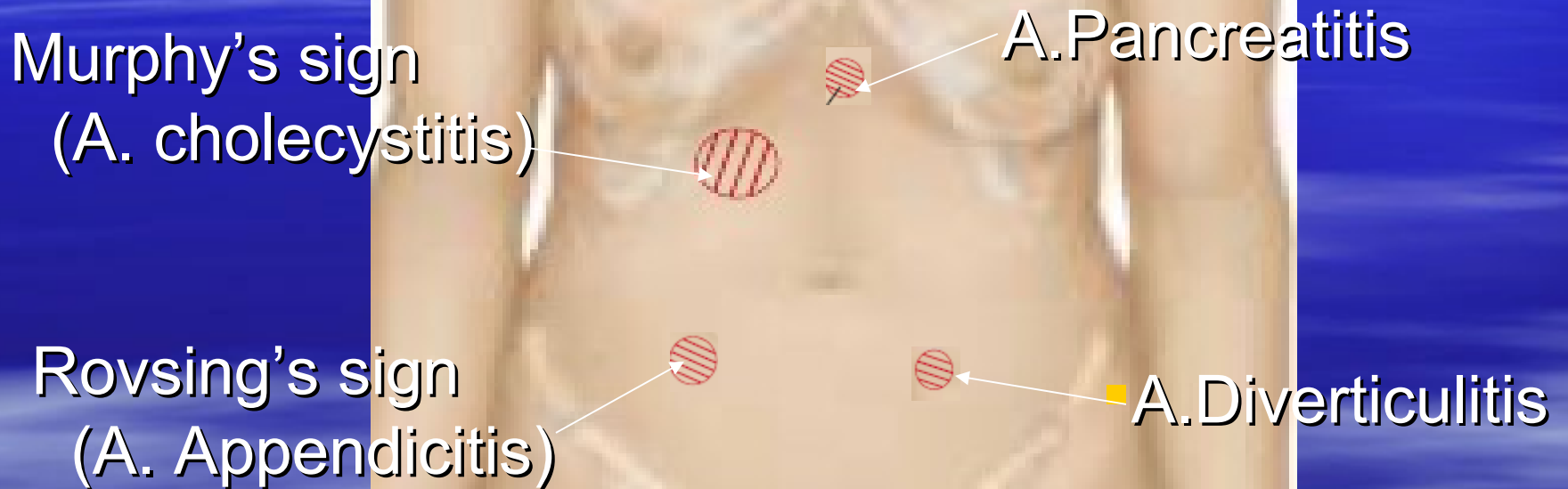
- **Purpose:** to detect enlarged organs , masses and rebound tenderness.
- **Techniques:-**
 - Relax abdominal wall muscles with the ff maneuvers.
 - instruct z patient to breath quietly .
 - Flex the knees
 - Distract the patient.
 - Performed single / Double handed.

- palpate placing flat part of hand on abdomen & exert pressure gently with finger tips.

Abnormal findings.

- **Rebound tenderness**:- - Elicited by deeply pressing & release & it's sign of peritoneal irritation
- RUQ deep & rebound tenderness (Murphy's sign) of A. cholecystitis
- LLQ deep & rebound tenderness (Rovsing's sign) of A. Appendicitis
- **Mass**:- Should be described for the following.
 - Site : based on topography.
 - Size & Consistency (soft, firm, hard)
 - Surface (smooth, rough) & tenderness
 - Boarder (regular/irregular or sharp/blunt)
 - Mobility. & Attachment to (over lying or underlying structures)

Sites of Diagnostic tenderness signs



3. Liver Palpation



- **Techniques:-** place left hand posteriorly b/n Rt 12th rib & iliac crest to press forward and Rt hand in RUQ .
- palpate while the patient inspires , starting from RLQ up to RUQ.
- Use hooking method in obese patient.
- Normally – Liver is firm, smooth surfaced & edge is palpable.
- Normal variation – enlarged Rt Lobe (Reidel lobe)

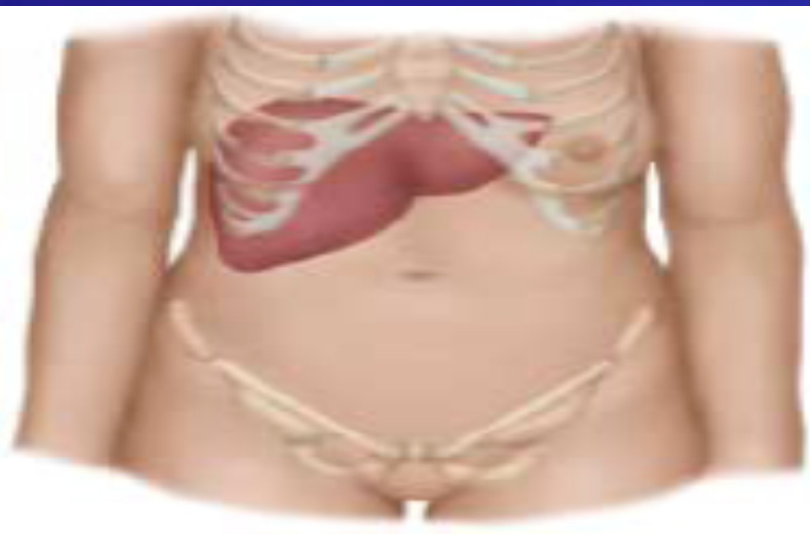


HEPATOMEGALY:- enlargement of Liver

Causes with characteristics Findings

➤ Causes of Palpable Tender Liver.

- Hepatitis
- Liver Abscess
- Congested Liver – due to RHF



➤ Causes of Hard palpable liver with rough surface & irregular boarder.

- Leukemia
- Lymphoma
- 1^o / 2^o Liver Ca s
- Liver TB



4. Spleen Palpation



- **Techniques:** Place Lt hand around the patient to support & press forward left rib cage & palpate with Rt hand from umbilicus towards LUQ (Bimanual palpation)
- **Splenomegally :** enlargement of spleen.
 - spleen should enlarge 2-3x to be palpable.
 - it enlarges downward & towards umbilicus.
 - Enlargement confirmed by:-
 - if can't go above it .
 - If medial notch is palpable.

5. Palpation of Kidneys



1. **Bimanual Palpation** – Normally not palpable

- Techniques: Place Lt/Rt hand around the patient to support & press forward left /Rt rib cage & palpate with Rt/ Lt hand.
- Causes of enlargement – Cysts, Tumors or Hydronephrosis.

2 **Assesing Kidney tenderness.**

- causes of + CVAT
- Pylonephritis. Renal stone .

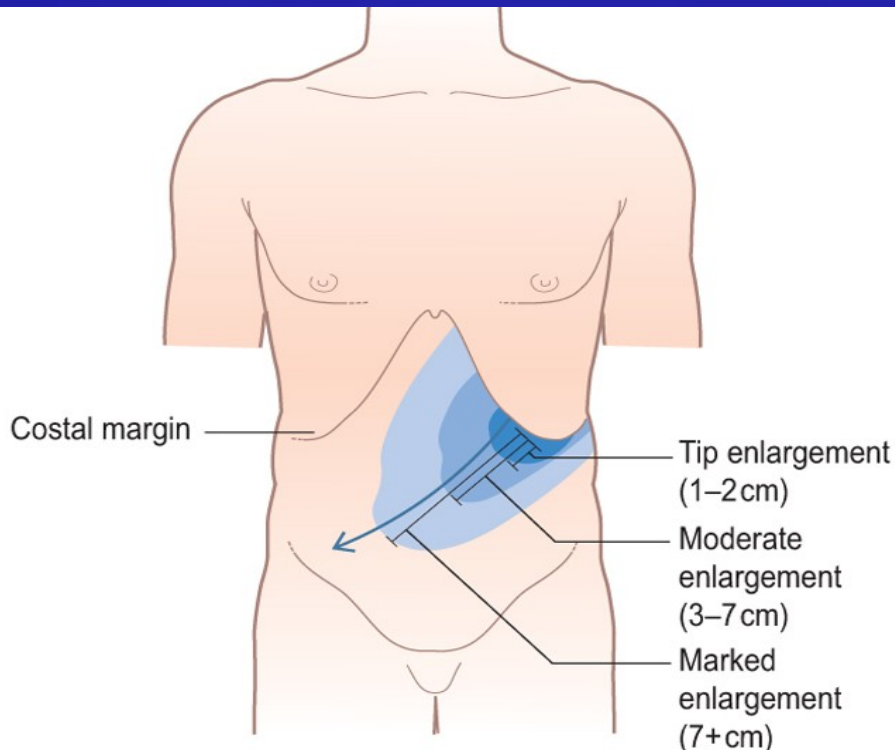


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■ Grades of Splenomegally

Grade	Enlargement
-------	-------------

- | | |
|------------|-------|
| - Tipped | 1-2cm |
| - Moderate | 3-8cm |
| - Huge | >8 cm |



■ Causes of splenomegally

1. Infections - Malaria, TB, Enteric fever, infective endocarditis.
2. Others- Leukemia, Lymphoma, portal hypertension
3. Causes of Huge splenomegally.
 - Chronic malaria
 - CML & Visceral Leshmania

How to Differentiate Spleen from Left Kidney ??

Spleen

- Palpable along line growth. (towards umbilicus)
- Can't go above it.
- Finger can't probe b/n spleen & Costal margine.
- Moves with respiration.
- Medial boarder palpable
- Notch Palpable
- Dull to percussion.

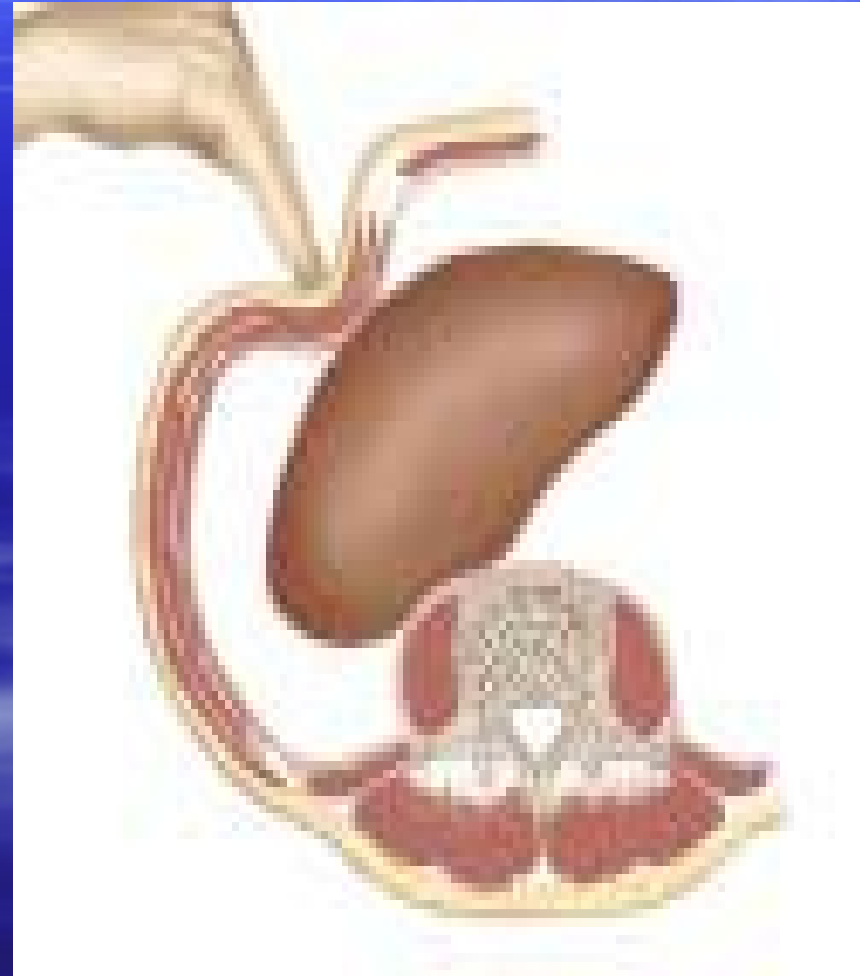
Left Kidney

- No
- Can go above it.
- Finger can probe b/n Lt Kidney & Costal margine
- No
- Not Much
- Not
- Tympanitic to Percussion.

6. Balloting (Identifying an organ or Mass in Ascitic Abdomen.

■ Techniques:-

- straighten &stiffen fingers
- Place on Abdomen
- Make A brief jabbing mov't
(this displaces the fluid)
- Try to feel surface of
enlarged mass/ organ.
- If felt – Say it is Ballot able

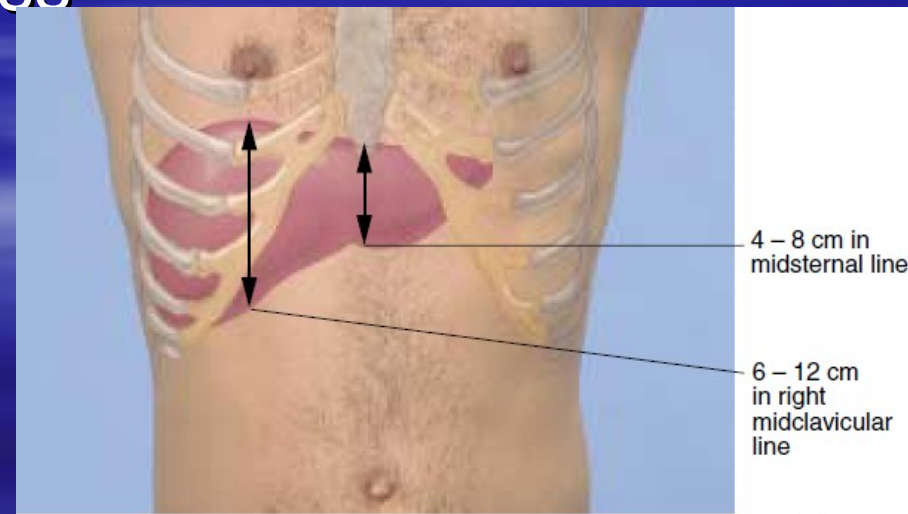
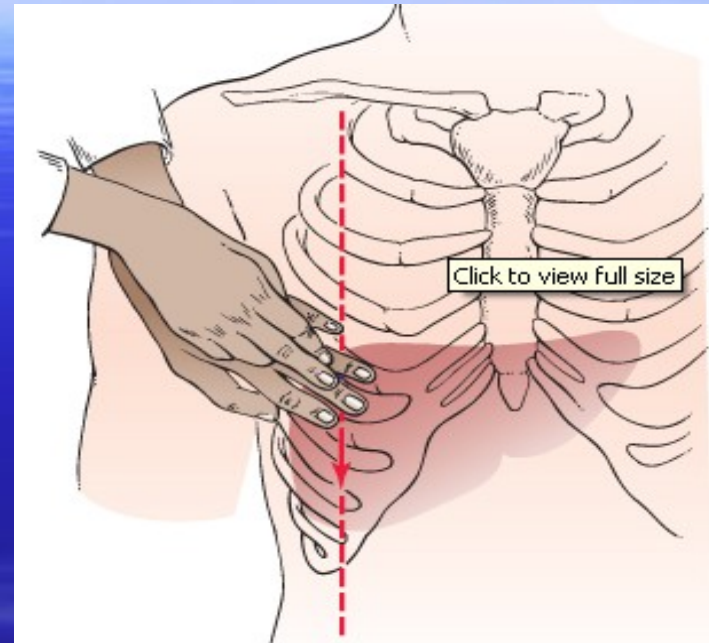


4.PERCUSSION

- **Purpose:-** - To assess amount & distribution of Gas.
 - To confirm Enlargement of mass or organ.
 - To estimate Size of Liver.
 - to detect Splenomegally.
 - To detect Presence of Intraperitoneal Fluid.
- **Technique:** percuss the Abdomen lightly in all 4 Quadrants.
 - Normally Tympanitic all over b/c of gas in GIT.
 - Abnormal findings.
 - Dull percussion Note in:- Enlarged mass or organ.
(Organomegally) & Intraperitoneal Fluid.
 - Hypertympanitic :- in intestinal obstruction.

1. Percuss Liver

- **Purpose:** to estimate liver size
- **Technique:** percuss in the right mid clavicular line from mid chest down to change in percussion note from dull to tympanitic.
- Measure this span of dullness it is called total vertical Liver Span. (TVLS)
- Normal 6-12cm
- Hepatomegally $< 12\text{cm}$



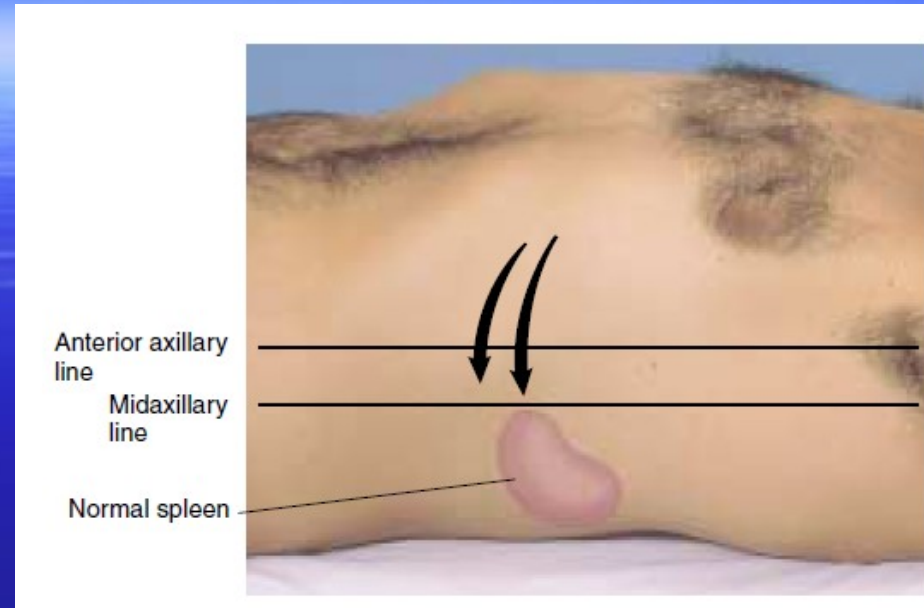
2.Percuss Spleen

A) Detect Splenomegally

- Technique : percuss in The lowest IC space Along left AAL.(Traube's space)
- Normal – resonant
- Splenomegally- Dull.

B) Check Splenic percussion sign.

- Percuss like above ,but after z patient take deep Breath.

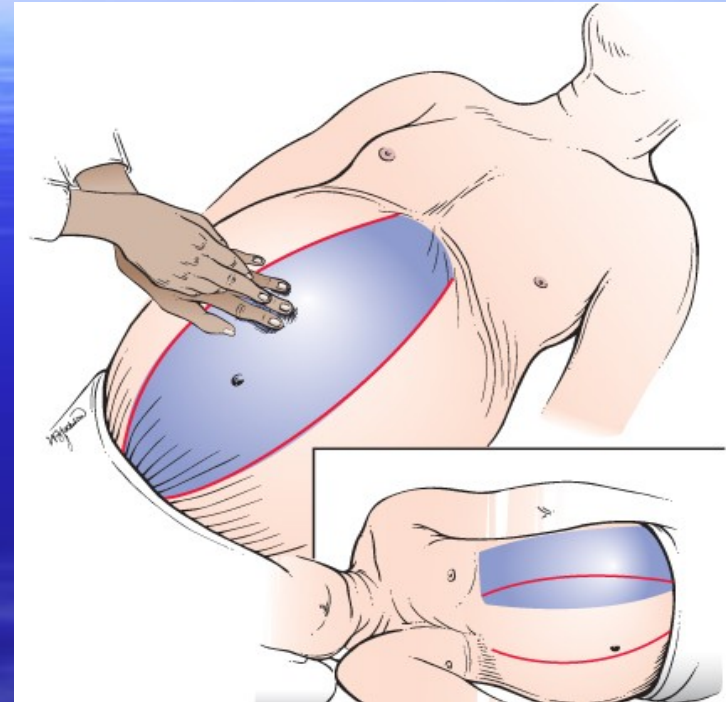


3. Detection of Intraperitoneal fluid collection

- using 2 Tests.

1. Shifting Dullness test

- Technique – percuss & determine boarder of Tympany & dullness
- Turn patient on His Side & Again percuss & determine boarder .



2 Fluid Thrill (Wave)

- Technique: press midline of Abdomen.
- Tap on one side & Feel Impulse on opposite side.

